MUKUL SHARMA

Machine Learning Scientist

LLM, RAG & Knowledge AI

Contact

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PROFILE

Machine Learning Scientist (M.Sc. Computer Science, grade 1,9) who builds **Retrieval-Augmented Generation (RAG)** systems: query processing, **hybrid and semantic retrieval**, **vector database (ChromaDB)** retrieval, **knowledge-graph retrieval**, **LLM-based relevance filtering**, and prompt-engineered answer generation through LLM APIs. Built a knowledge graph-based RAG system for open-domain question answering (M.Sc. thesis) and is currently building a RAG assistant that acts as first-level support on websites. Brings **4+ years of professional software engineering** at HCL Technologies and Adobe (Python, C, Git, CI/CD, Docker, Google Cloud) to LLM application development.

CORE COMPETENCIES

RAG & Retrieval	Retrieval-Augmented Generation (RAG), query processing, document chunking, hybrid retrieval, semantic search, embedding-based similarity search, retrieval strategies, knowledge-graph retrieval, LLM-based relevance filtering (critique)
LLMs & Prompting	LLM API integration, prompt engineering, chain-of-thought prompting, knowledge-grounded generation, Transformers, tokenization
Vector Databases	ChromaDB: chunk storage and vector-database retrieval
Software Engineering	Python, JavaScript, HTML, CSS, C; Git, testing, debugging, code review, documentation; CI/CD with Jenkins
Cloud & Infrastructure	Google Cloud Platform (GCP), Docker, Linux, HPC with SLURM
ML & Data	PyTorch, scikit-learn, Pandas, NumPy; CNN, RNN, LSTM, GAN; ETL pipelines, data processing and augmentation
Computer Vision	OCR, YOLO object detection, image segmentation, OpenCV

RAG & LLM PROJECTS

RAG-Based Website Support Assistant (First-Level Support) — Ongoing

Personal project

- Building a RAG system that acts as the **first point of contact for website visitors**, answering their questions and providing first-level support.
- Implemented **hybrid retrieval** to find the most relevant content for each user query.
- Added an **LLM-based critique step** that filters out irrelevant retrieved items, so only relevant context reaches answer generation.
- Integrated **LLM API calls** and applied **chain-of-thought prompt engineering** to generate clear answers grounded in the retrieved context.

Knowledge Graph-Based RAG for Open-Domain Question Answering — M.Sc. Thesis

Leibniz University Hannover | Reasoning-augmented generation with a knowledge graph as retriever

- Built a **RAG system that uses a knowledge graph as the retriever** for reasoning-augmented generation in open-domain question answering.
- Implemented the retrieval pipeline: **query processing**, **semantic search**, and **vector-database retrieval**.
- Stored text chunks in **ChromaDB** and applied multiple **retrieval strategies** to supply relevant evidence for answer generation.
- Researched **trustworthy, knowledge-grounded generation** aimed at reliable answers.

PROFESSIONAL EXPERIENCE

Research Assistant (Student) 03/2025 – 06/2026
Institute of Manufacturing Engineering and Machine Tools, Hannover, Germany

- Contributed to research projects on **automated recognition and extraction of component information** from technical drawings and 3D data.
- Developed **Python-based machine learning and computer vision pipelines**, including **OCR-based information extraction** and **YOLO-based object and symbol detection**.
- Prepared, structured, and **augmented datasets** with OpenCV; **trained models on HPC clusters** to improve performance and inference quality.

Software Developer 03/2023 – 03/2024
Adobe, India

- Designed and implemented small functional features in **C** for Adobe software.

Senior Analyst (Software) 01/2020 – 02/2023
HCL Technologies, India

- Analyzed requirements and supported the **software development lifecycle (SDLC)** from design through deployment.
- Developed software components in **Python, JavaScript, HTML, and CSS**; applied **testing, debugging, documentation, and code review** to ensure software quality.
- Built automated **ETL pipelines** for data ingestion, cleaning, transformation, and preprocessing.
- Developed **machine learning models** for predictive analysis and trend forecasting.
- Implemented **CI/CD workflows with Git and Jenkins**; containerized applications with **Docker** and deployed solutions on **Google Cloud Platform**.
- Automated data-processing and deployment workflows to improve efficiency, reliability, and maintainability.

EDUCATION

M.Sc. Computer Science 04/2024 – 06/2026
Leibniz University Hannover, Germany | Final grade: 1,9

- Focus: Machine Learning, Deep Learning, Natural Language Processing. Thesis: see RAG & LLM Projects.
- Designed and ran ML experiments for classification and regression; built models from scratch (Linear and Logistic Regression, CNNs, RNNs, GANs).
- Trained and evaluated models on HPC clusters using SLURM; performed model optimization, hyperparameter tuning, and performance evaluation.
- Wrote research reports and technical documentation covering methods, experiments, and results.

B.Tech. Computer Science 08/2015 – 10/2019
Dr. A.P.J. Abdul Kalam Technical University, India | Final grade: 78/100 (approx. German grade 2,3)

- Thesis: University Recommendation System. Focus: Data Structures and Python.

CERTIFICATES

Associate Cloud Engineer | Google Certified | 2022
Cloud Digital Leader | Google Certified | 2023
Microsoft 365 Fundamentals | Microsoft Certified | 2022

LANGUAGES

English: Fluent (C1) | **German**: Basic (A2)